

DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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Abstract—Physics-based humanoid motion imitation requires simultaneously tracking multiple objectives, including root motion, body pose, and velocities. Existing methods often rely on handcrafted reward terms with manually tuned weights and scales, which can require substantial adjustment across different motions. We propose DARE (Differential Adversarial Reward Equalization), an adversarially learned reward method based on the differential between the reference motion and the policy state. DARE represents perfect tracking as a zero differential and learns a unified tracking reward from policy-generated residuals, while controlling discriminator sensitivity and automatically calibrating its output scale to support a consistent reward configuration across motions. We evaluate DARE on multiple humanoid skills with diverse dynamics and contact patterns. The results show that DARE achieves stable and accurate motion tracking without motion-specific reward tuning.

I. INTRODUCTION

Physics-based humanoid motion imitation requires the controller to reproduce a prescribed reference motion under dynamic constraints, actuator limits, and complex contact conditions. Whether an imitation succeeds is not a single geometric question. The root must translate and turn as the reference does, the articulated body must maintain consistent poses, joint and root velocities must stay synchronized with the reference motion, and contacts must occur at the right moments. Because these objectives have different physical units, numerical ranges, and dynamical characteristics, and are coupled through the robot dynamics, motion imitation is inherently a multi-objective control problem.

Traditional methods transform several tracking errors separately and combine them into a scalar reward through manually chosen weights and scales. These coefficients determine both the relative importance and the numerical response of the tracking objectives. Because typical errors and dynamical difficulty change with the motion, a coefficient set that serves one skill need not serve another; in practice, reward design is often re-tuned whenever the reference motion changes.

Adversarial imitation offers a way to reduce manual reward design. Instead of declaring in advance how many meters of root-position error are worth how many radians of rotation error, a discriminator learns a unified imitation signal directly from data. AMP and related methods learn rewards that describe motion style and plausibility by discriminating whether short state transitions come from the reference distribution [5]. For motion tracking that must strictly follow a synchronized reference trajectory, ADD further feeds the reference-policy state differential to the discriminator and asks whether it is close to the ideal zero differential [6]. The aggregation among tracking objectives can then be learned

by the discriminator instead of specifying component-wise weights and scales by hand.

Differential adversarial rewards, however, still face an important issue. Their ideal sample is always a single point, the zero differential. Without a suitable constraint, the discriminator easily becomes too sharp: it assigns a high score to the origin and compresses the many nonzero residuals produced by the policy into a nearby low-score region. Even if classification succeeds, the discriminator may not provide an effective continuous learning signal for the policy. Existing methods constrain the discriminator through regularization, but one regularizer simultaneously affects two distinct properties. On the one hand, it controls the discriminator’s local sensitivity, that is, how fast the score changes with the input residual. On the other hand, it also affects the numerical range in which the final scores live, and this range in turn determines whether the policy reward operates in an effective interval. As the residual distribution changes across motions, the same regularization setting can correspond to different sensitivities and different score scales, producing motion-dependent effective regularization requirements.

The core idea of DARE is to treat these two issues separately rather than control them with one motion-dependent coefficient. First, DARE applies spectral normalization to every linear layer of the differential discriminator, directly bounding the maximum amplification of residual variations. Second, after residual normalization is fixed, DARE measures the score separation between the unique zero-error anchor and representative policy residuals and rescales the discriminator once in closed form. The geometry constraint and reward operating scale are thus specified independently.

Constraining the critic’s sensitivity alone does not resolve the score scale. A Lipschitz bound controls how rapidly the score may change, but not the numerical range actually used by the trained critic. DARE therefore measures the anchor-to-typical-residual score gap once after normalization stabilizes and uses a closed-form global rescaling tied to the response range of the reward transform. All motions use exactly the same calibration rule; the only quantity that changes across motions is the residual statistic measured automatically from the training data. The critic’s geometry and score scale thus no longer depend on the same hand-tuned parameter.

The main contributions are threefold. First, we identify that the regularizer in existing differential adversarial rewards simultaneously affects the critic’s local geometry and its score scale, and that this coupling can make different motions require different effective regularization settings. Second, we propose DARE, which constrains discriminator sensitivity

through layer-wise spectral normalization and sets the score scale automatically through one-shot anchor calibration, removing the need for a motion-specific gradient-penalty coefficient. Third, we evaluate one shared DARE configuration on six skills with different durations, contact patterns, and dynamical characteristics.

II. RELATED WORK

Physics-based motion tracking and learned imitation objectives. DeepMimic combines pose, velocity, root, and end-effector tracking terms for physics-based skills [1]; subsequent work improves scalability and motion coverage through architectures, representations, and data [2], [3]. Adversarial imitation provides a learned alternative to a manually written reward: GAIL matches expert and policy occupancies [4], while AMP learns a motion prior by discriminating short state transitions [5]. ADD addresses reference-conditioned tracking by presenting the reference-policy differential to an adversarial discriminator and learning a nonlinear aggregation of tracking objectives without component-wise reward weights [6].

Multi-objective learning and objective aggregation. Competing objectives are commonly combined by weighted scalarization, Pareto optimization, or adaptive loss weighting [10], [11]. These approaches differ in whether preferences are specified explicitly or adapted from objective statistics. In differential adversarial tracking, the discriminator instead learns this aggregation from the joint residual distribution.

Regularization of adversarial critics. Adversarial critics are regularized with gradient penalties or spectral normalization to control sensitivity and improve optimization [7], [8]. Gradient penalties constrain sampled input derivatives, whereas spectral normalization bounds the gain of individual network maps by construction. For non-Wasserstein adversarial losses, the numerical score range also determines the operating regime of the loss and its reward transform [9]. Geometry and score calibration are therefore related but distinct considerations when a critic supplies the learning signal.

III. BACKGROUND

A. Reference-Conditioned Motion Tracking

In reference-conditioned humanoid motion imitation, the controller must keep the executed motion continuously aligned with a time-synchronized reference trajectory. Let s_t and $\hat{s}_t$ denote the current state and the reference state at time t . Tracking quality is jointly measured by several physical quantities, including root position and orientation, articulated body pose, and motion velocities. Let $\phi_i(s_t)$ extract the i -th tracking quantity from the current state and let $\phi_i(\hat{s}_t)$ denote its reference counterpart. These quantities may have different dimensions and physical units—positions are measured in meters, while rotations and velocities have their own representations.

B. Handcrafted Objective Scalarization

Traditional methods first compress each tracking error into a scalar. For tracked quantity i , the scalar tracking error at time t is the Euclidean distance $e_{i,t} = \|\phi_i(\hat{s}_t) - \phi_i(s_t)\|_2$.

A representative handcrafted scalarization is

$$r_t^{\text{hand}} = \sum_i w_i \exp(-\alpha_i e_{i,t}^2), \quad (1)$$

where w_i and α_i set the relative importance and numerical scale of each tracking quantity. DeepMimic is a representative instance of this class: its tracking terms use weights and error scales specified before training and held fixed throughout [1]. This is a fixed scalarization of the tracking objectives: the trade-off among errors is fixed through constant coefficients chosen before training starts. Because typical error magnitudes and dynamical characteristics differ across motions, one coefficient set need not serve all motions. Learned scalarization instead leaves the trade-off unspecified in advance and learns the aggregation among objectives from the error distribution that actually occurs during training.

C. Differential Adversarial Objectives

Differential adversarial learning provides such a learned aggregation [6]. Consider n non-negative objective losses $l^1, \dots, l^n$ collected into a differential vector $\Delta = (l^1, \dots, l^n)$, where each entry records the gap between the current solution and the ideal on one objective. When all objectives are simultaneously ideal, $\Delta = 0$. A discriminator is trained to assign a high score to the ideal zero differential and lower scores to the nonzero differentials produced by the current policy. Because it receives all objective differentials jointly, the discriminator can learn a nonlinear aggregation of the objectives without fixed per-objective weights. Its sensitivity to individual objectives may vary across the residual space and throughout training, allowing the learned aggregation to adapt to the differential distribution encountered during optimization.

In synchronized motion tracking, the differential can be formed directly from the feature residual between the current and reference states rather than from one scalar loss per objective. Perfect tracking again corresponds to the unique zero differential, while nonzero differentials preserve the magnitude, direction, and correlations of the tracking errors. The discriminator score can therefore be used to construct a policy reward that increases as the current differential is judged more consistent with perfect tracking.

IV. DARE: DIFFERENTIAL ADVERSARIAL REWARD EQUALIZATION

A. Learned Differential Reward

Motion tracking involves multiple objectives with different physical meanings. Rather than compressing them into separately weighted scalar terms, DARE forms one full reference-policy residual. Let $\phi(s_t) \in \mathbb{R}^d$ collect the tracked quantities and let $\nu \in \mathbb{R}^d$ contain their per-coordinate mean-absolute scales, estimated during early training and fixed after the first 10^8 samples. The normalized differential is $\Delta_t = \delta_t \oslash \nu$,

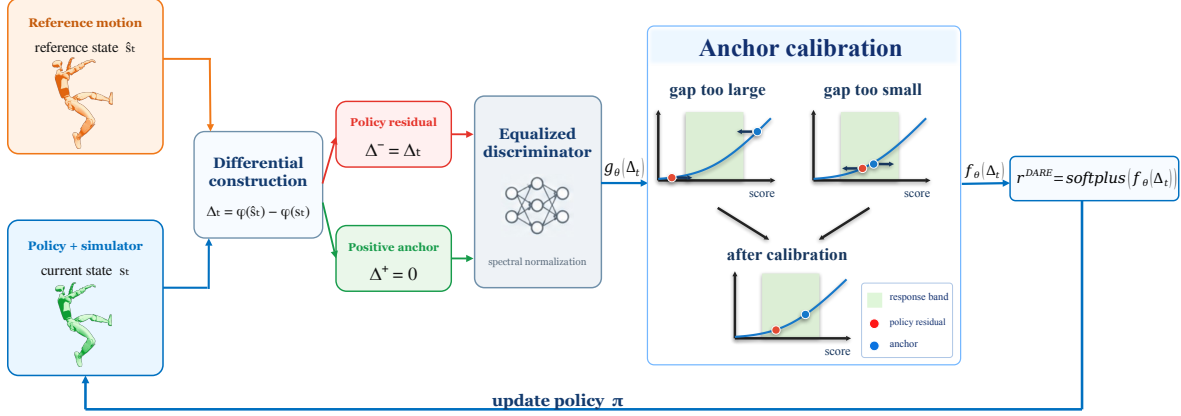


Fig. 1. DARE training loop. The synchronized reference and simulated state form the policy residual. The residual and zero anchor train the equalized discriminator; one-shot anchor calibration rescales its logit, and the resulting softplus reward drives the PPO update.

where $\delta_t = \phi(\hat{s}_t) - \phi(s_t)$, and perfect tracking is represented by $\Delta^+ = 0$. Here $\oslash$ denotes element-wise division. Because normalization only rescales coordinates and introduces no offset, perfect tracking remains the unique zero differential. Rotational quantities use a tangent-normal representation so that all tracked residuals are represented in consistent Euclidean coordinates.

The full differential retains coordinate-wise error magnitudes, directions, and correlations. Two states can therefore have the same aggregate distance while differing in which tracked quantities deviate from the reference. Let q_π denote the distribution of differentials induced by policy π . During training, recently generated policy differentials are stored in a replay buffer and reused as negative samples for discriminator updates.

DARE maps the full differential to a scalar discriminator logit, $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$. A larger $f_\theta(\Delta)$ indicates that the differential is judged more consistent with perfect tracking. We introduce a global logit scale $\kappa > 0$, initialized to 1 and determined automatically once by the anchor calibration described in Sec. IV-C. The scaled logit used for discriminator training and policy reward is therefore $\kappa f_\theta(\Delta)$.

Perfect tracking defines the positive differential $\Delta^+ = 0$, while differentials induced by policy π form the negative class, with $\Delta^- \sim q_\pi$. The discriminator is trained with the balanced binary logistic loss

$$\mathcal{L}_{\text{DARE}}(\theta) = \frac{1}{2} \text{softplus}[-\kappa f_\theta(\Delta^+)] + \frac{1}{2} \mathbb{E}_{\Delta^- \sim q_\pi} \text{softplus}[\kappa f_\theta(\Delta^-)]. \quad (2)$$

The first term encourages a large logit for the zero differential, while the second encourages lower logits for policy-induced residuals. Because the discriminator receives

all residual coordinates jointly, it learns from data how the different tracking errors should contribute to a single score.

The policy reward is constructed from the same scaled logit:

$$r_t^{\text{DARE}} = \text{softplus}[\kappa f_\theta(\Delta_t)]. \quad (3)$$

Since softplus is strictly increasing, a larger discriminator logit produces a larger policy reward. The learned reward therefore encourages the policy toward residuals that the discriminator judges more consistent with perfect tracking, while jointly accounting for root motion, body pose, and velocities.

B. Equalized Discriminator: Sensitivity Constrained by Construction

The differential discriminator must not only distinguish perfect tracking from policy-induced residuals, but also provide the policy with a continuous reward signal that reflects differences in tracking quality. Its sensitivity to the tracking differential therefore needs to be controlled. Since perfect tracking always corresponds to the unique zero differential $\Delta^+ = 0$, excessive sensitivity near the origin can cause the discriminator logit to decrease sharply even for a small deviation from zero. In this case, although the discriminator may still separate positive and negative samples, nonzero residuals of different quality can be compressed into a similar low-score region, weakening the ability of the reward to distinguish tracking quality.

A common approach is to introduce a gradient penalty during discriminator training to limit excessive input sensitivity [6]. However, this introduces an additional regularization strength: if the constraint is too weak, the discriminator may remain overly sensitive; if it is too strong, the discriminator may become overly smooth and lose its ability to distinguish different residuals. DARE instead constrains the maximum

response of the discriminator directly through its network structure, without introducing an additional balancing coefficient.

The normalized full tracking differential $\Delta \in \mathbb{R}^d$ is fed directly into the discriminator. Let the network contain L affine layers, with weight matrix W_ℓ and bias b_ℓ at layer ℓ . DARE applies spectral normalization to every weight matrix,

$$\bar{W}_\ell = \frac{W_\ell}{\sigma_{\max}(W_\ell)}.$$

Here, $\sigma_{\max}(W_\ell)$ is the largest singular value of W_ℓ , defined as

$$\sigma_{\max}(W_\ell) = \max_{v \neq 0} \frac{\|W_\ell v\|_2}{\|v\|_2}.$$

It therefore gives the maximum factor by which the linear map can amplify an input variation. Consequently, for any vector v ,

$$\|\bar{W}_\ell v\|_2 \leq \sigma_{\max}(W_\ell) \|v\|_2.$$

After spectral normalization, $\sigma_{\max}(\bar{W}_\ell) = 1$. The network uses ReLU as the activation function ρ , which is also 1-Lipschitz and therefore does not increase the Euclidean distance between two inputs.

Consider an arbitrary hidden layer ℓ , and let x and x' denote two inputs to this layer. Its outputs satisfy

$$\begin{aligned} & \|\rho(\bar{W}_\ell x + b_\ell) - \rho(\bar{W}_\ell x' + b_\ell)\|_2 \\ & \leq \|\bar{W}_\ell(x - x')\|_2 \\ & \leq \sigma_{\max}(\bar{W}_\ell) \|x - x'\|_2 = \|x - x'\|_2. \end{aligned} \quad (4)$$

The first inequality follows from the 1-Lipschitz property of ReLU; the common bias b_ℓ cancels when the two pre-activations are compared. The second follows directly from the largest-singular-value bound above, and the final equality follows from $\sigma_{\max}(\bar{W}_\ell) = 1$. Thus, every hidden layer is non-expansive. The final affine output layer obeys the same bound without the ReLU activation. Applying this relation successively through all layers therefore gives

$$|f_\theta(\Delta) - f_\theta(\Delta')| \leq \|\Delta - \Delta'\|_2. \quad (5)$$

Thus, a small change in the tracking differential cannot be arbitrarily amplified into a large change in the discriminator logit; its maximum sensitivity is directly constrained by the network structure.

Since perfect tracking corresponds to the zero differential $\Delta^+ = 0$, setting $\Delta' = 0$ in Eq. (5) gives

$$|f_\theta(\Delta) - f_\theta(0)| \leq \|\Delta\|_2. \quad (6)$$

Therefore, when the tracking differential is close to the perfect-tracking anchor, the discriminator logit cannot change arbitrarily rapidly relative to its value at zero residual. DARE thereby constrains the maximum response of the discriminator to tracking residuals through layer-wise spectral normalization.

C. Anchor Calibration: Closed-Form Output Scale

Spectral normalization constrains how rapidly the raw discriminator logit $f_\theta(\Delta)$ can change with the tracking differential, but it does not determine the global numerical scale of the logit used by the reward. DARE controls this remaining degree of freedom through the global scale κ .

For a scalar logit z , the slope of the softplus reward is

$$\frac{d}{dz} \text{softplus}(z) = \sigma(z). \quad (7)$$

When z is sufficiently negative, $\sigma(z)$ is close to zero, so changes in the logit produce only small reward differences. Around $z = 0$, the reward becomes more responsive, and as z increases the slope approaches one. DARE therefore uses the interval over which the softplus slope increases from 0.2 to 0.8 as its target response region. Its width is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 16 \approx 2.77. \quad (8)$$

The target width w specifies the desired separation in scaled-logit space. To determine the corresponding scale automatically, DARE measures the separation produced by the discriminator itself. At calibration time t_f , let

$$M_f = f_\theta(0) - \frac{1}{2} \left[\mathbb{E}_{\Delta \sim q_{\text{cur}}} f_\theta(\Delta) + \mathbb{E}_{\Delta \sim q_{\text{rep}}} f_\theta(\Delta) \right] \Bigg|_{t=t_f}, \quad (9)$$

where q_{cur} and q_{rep} denote the residual differential distributions from the current rollout and replay buffer, respectively. Thus, M_f is the raw-logit separation between the zero-differential anchor and the policy residuals observed at calibration.

DARE then sets the global logit scale in closed form:

$$\kappa = \frac{w}{M_f}. \quad (10)$$

The measured separation is therefore rescaled from M_f to $\kappa M_f = w$. If $M_f > w$, then $\kappa < 1$ and an overly wide separation is compressed; if $M_f < w$, then $\kappa > 1$ and an overly narrow separation is enlarged. Since the same positive factor multiplies every discriminator logit, this operation preserves their ordering and only changes the global output scale.

Calibration is performed once after the input normalization is fixed. Before calibration, $\kappa = 1$. After Eq. (10) is evaluated, κ is kept fixed and the scaled logit $\kappa f_\theta(\Delta)$ is used consistently in both discriminator optimization and policy reward computation. Spectral normalization therefore controls the sensitivity of the raw discriminator mapping, while anchor calibration independently sets the global scale at which that mapping is used by the reward.

D. Training

Training alternates rollout collection, differential construction, discriminator updates, and PPO updates (Algorithm 1). Discriminator minibatches use policy residuals from both the current rollout and replay buffer together with the zero differential as the positive class. At t_f , before the rewards

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_ϕ , value V_ω , discriminator f_θ , replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- 1: **for** iteration $t = 1, 2, \dots$ **do**
- 2: Collect a synchronized rollout with π_ϕ from random reference phases; construct $\Delta_t = \delta_t \odot v$; add to $\mathcal{B}$
- 3: **if** $\mathcal{N}$ was fixed this iteration **then**
- 4: $\bar{f}_{\text{cur}} \leftarrow \mathbb{E}_{\Delta \sim q_{\text{cur}}} f_\theta(\Delta)$
- 5: $\bar{f}_{\text{rep}} \leftarrow \mathbb{E}_{\Delta \sim q_{\text{rep}}} f_\theta(\Delta)$
- 6: $M_f \leftarrow f_\theta(0) - \frac{1}{2}(\bar{f}_{\text{cur}} + \bar{f}_{\text{rep}})$; $\kappa \leftarrow \ln 16/M_f$
- 7: **end if**
- 8: Compute r_t^{DARE} by (3)
- 9: **for** discriminator minibatch **do**
- 10: Sample current and replay differentials and use $\Delta^+ = 0$ as the positive anchor
- 11: Update f_θ by one step on (2)
- 12: **end for**
- 13: Update π_ϕ and V_ω with PPO on r_t^{DARE}
- 14: **end for**

TABLE I

DARE CONFIGURATION SHARED ACROSS ALL SIX SKILLS. THE CALIBRATION RULE IS SHARED; THE RESULTING κ IS DATA-DEPENDENT.

Component	Setting
Differential input	single frame, $d = 172$
Discriminator	MLP, $172 \rightarrow 1024 \rightarrow 512 \rightarrow 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Logit calibration	$\kappa = \ln 16/M_f$, once at iter. 382
Critic optimizer	SGD, lr 2.5×10^{-4} , momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials

of that iteration are computed, M_f is measured with the discriminator in evaluation mode and κ is set by Eq. (10). One discriminator, one reward, one policy, and one calibration rule make up the complete training loop.

V. EXPERIMENTS

A. Experimental Setup

All methods are evaluated using the same 28-DoF, 15-body humanoid model adopted in prior physics-based motion-imitation studies, simulated in Isaac Lab at 120Hz with actions applied at 30Hz through PD joint-position control [14], [15]. All four methods share the same environment, training budget, motion clips, random-phase initialization, and evaluation protocol. Each policy is trained for 2000 iterations of $8192 \text{ environments} \times 32 \text{ steps}$ (5.24×10^8 samples) with five independent seeds, and is evaluated on 4096 episodes per seed from randomly sampled reference phases. Actor and value networks are two-layer MLPs (1024, 512), and the PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$, SGD with momentum 0.9) are fixed across methods. DARE additionally uses the normalizer and one-shot logit calibration specified in Table I.

The benchmark contains six skills with substantially different motion durations, contact patterns, and dynamical characteristics: Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick. Training termination follows each method’s environment configuration (pose termination is enabled for DeepMimic, ADD, and DARE and disabled for AMP). The evaluation harness accumulates each metric over the executed reference-aligned steps and uses the same aggregation rule for every method; terminated trajectories are not padded with synthetic states. Final policies reach essentially the full prescribed evaluation horizon, so early termination has negligible effect on the reported averages.

We use six complementary measures. The four geometric measures quantify frame-wise imitation fidelity, whereas foot progress and lateral drift capture trajectory-level behavior that is not summarized by instantaneous pose errors. Four describe geometric tracking at two spatial levels: root position and root rotation measure global translation and orientation, while body position and body rotation measure articulated-body tracking. For an evaluation trajectory with B bodies, where r, p_b and q, q_b denote simulated root, body positions and orientations and hats denote reference values, these quantities are

$$\begin{aligned}
 E_{\text{root-pos}} &= \|\hat{r} - r\|_2, & E_{\text{root-rot}} &= \angle(q, \hat{q}), \\
 E_{\text{body-pos}} &= \frac{1}{B} \sum_{b=1}^B \|(\hat{p}_b - \hat{r}) - (p_b - r)\|_2, \\
 E_{\text{body-rot}} &= \frac{1}{B} \sum_{b=1}^B \angle(q_b, \hat{q}_b).
 \end{aligned} \tag{11}$$

where $\angle(\cdot, \cdot)$ is the shortest angle between normalized quaternions. Geometric quantities are evaluated at each reference-aligned timestep and then averaged over executed timesteps and episodes; the two behavior measures are computed per trajectory and then averaged over episodes. The remaining two measures describe motion-level behavior. Let Δr and $\Delta \hat{r}$ be the simulated and reference root displacements over the evaluated trajectory, and let $\Delta p_j, \Delta \hat{p}_j$ be the corresponding displacements of foot j . The reference-projected progress and transverse drift are

$$\begin{aligned}
 \rho_{\text{foot}} &= \min_j \frac{\Delta p_j^\top \Delta \hat{p}_j}{\|\Delta \hat{p}_j\|_2^2}, \\
 \rho_r &= \frac{\Delta r^\top \Delta \hat{r}}{\|\Delta \hat{r}\|_2^2}, \\
 E_{\text{lat}} &= \frac{\|\Delta r - \rho_r \Delta \hat{r}\|_2}{\|\Delta \hat{r}\|_2}.
 \end{aligned} \tag{12}$$

Foot progress is ideal at one, while lateral drift is minimized at zero. All six quantities are averaged over the evaluation trajectories; lower is better except for foot progress, whose target value is one.

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and Fig. 2 shows the resulting controllers executing the six skills in Isaac Lab. With one configuration shared across all six

TABLE II

MACRO-AVERAGE TRACKING AND MOTION-LEVEL BEHAVIOR UNDER THE MATCHED TRAINING AND EVALUATION PROTOCOL. FOR EACH SEED, EACH METRIC IS MACRO-AVERAGED EQUALLY OVER THE SIX SKILLS; ENTRIES REPORT THE MEAN $\pm$ ONE STANDARD DEVIATION ACROSS FIVE SEEDS. LOWER IS BETTER EXCEPT FOR FOOT PROGRESS, WHICH IS IDEAL AT ONE. DARE USES THE SAME CONFIGURATION ON ALL SKILLS.

(a) Geometric tracking				
Metric $\downarrow$	AMP	DeepMimic	ADD	DARE
Root Pos. [m]	0.9116 $\pm$ 0.084	0.0703 $\pm$ 0.012	0.0725 $\pm$ 0.010	0.0528$\pm$0.008
Body Pos. [m]	0.2169 $\pm$ 0.031	0.0557 $\pm$ 0.009	0.0363 $\pm$ 0.007	0.0313$\pm$0.006
Root Rot. [rad]	0.7311 $\pm$ 0.076	0.1459 $\pm$ 0.021	0.0673 $\pm$ 0.011	0.0529$\pm$0.009
Body Rot. [rad]	0.8098 $\pm$ 0.069	0.2145 $\pm$ 0.028	0.1358 $\pm$ 0.017	0.1246$\pm$0.013
(b) Motion-level behavior				
Metric	AMP	DeepMimic	ADD	DARE
Foot progress $\rightarrow$ 1	1.1335 $\pm$ 0.046	1.0060 $\pm$ 0.018	1.0109 $\pm$ 0.021	0.9989$\pm$0.012
Lateral drift $\downarrow$	0.1127 $\pm$ 0.019	0.0277 $\pm$ 0.007	0.0435 $\pm$ 0.009	0.0171$\pm$0.005

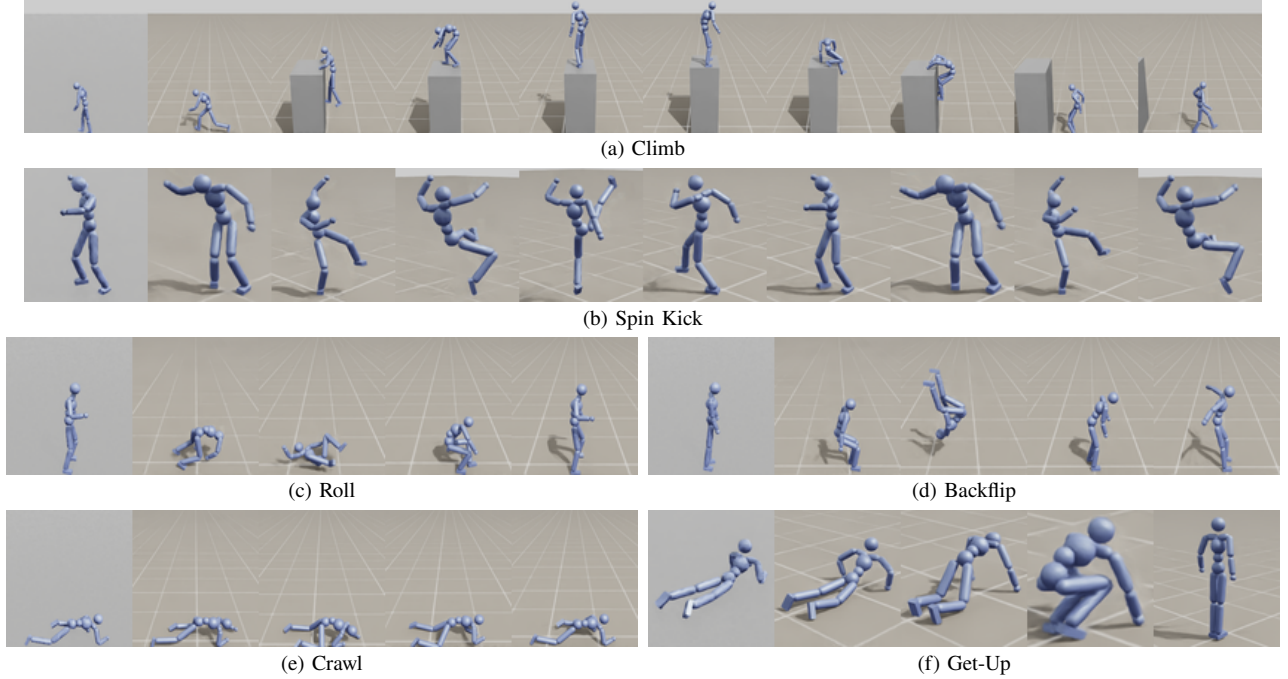


Fig. 2. DARE policies executing the six skills in Isaac Lab using the shared configuration in Table I. Frames are uniformly sampled from each rollout and rendered from a fixed front-view camera. Climb and Spin Kick are shown with ten frames, and the remaining skills with five.

skills, DARE achieves the best macro-average on all six measures. It yields the lowest root- and body-level position and rotation errors, the foot-progress ratio closest to one, and the lowest lateral drift. Relative to the strongest baseline for each measure, root-position error, root-rotation error, and lateral drift are reduced by approximately 25%, 21%, and 38%.

C. Learning Dynamics

Figure 3 compares the five-seed mean body-position and root-rotation errors against environment samples for Roll and Climb, which provide contrasting motion durations and contact structures. Across both, the DARE curves converge to lower errors under the same sample budget, complementing the final-checkpoint comparison in Table II.

D. Component Analysis

We isolate the two components that define DARE’s critic: layer-wise spectral normalization and one-shot score calibration. This gives a 2×2 comparison while keeping the differential input, binary objective, replay procedure, PPO budget, and evaluation protocol fixed. The study covers eight motions—Roll, Climb, Backflip, Crawl, Get-Up, Spin Kick, Push Box, and Sit on Sofa—with three independent seeds for every configuration. Entries are macro-averaged over the eight motions for each seed and then summarized as a mean $\pm$ standard deviation; cells whose measurements are still pending are marked in red.

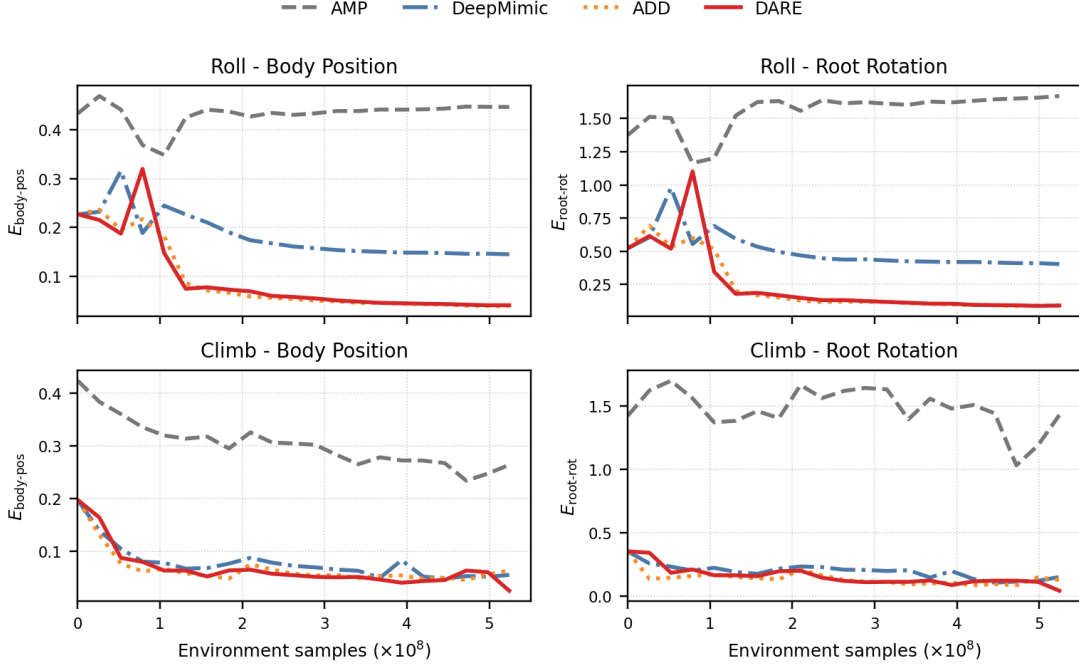


Fig. 3. Learning curves for Roll and Climb under the matched training budget. Curves show the mean over five independent seeds. The four panels report body-position and root-rotation errors versus environment samples, representing articulated translational fidelity and global orientation tracking, respectively. Lower is better.

TABLE III

COMPONENT ANALYSIS OVER EIGHT MOTIONS. SN AND CAL. DENOTE LAYER-WISE SPECTRAL NORMALIZATION AND ONE-SHOT CALIBRATION. LOWER IS BETTER; RED ENTRIES ARE PENDING MEASUREMENTS.

Method	SN	Cal.	R.Pos. ↓	B.Pos. ↓	R.Rot. ↓	B.Rot. ↓	DoF Vel. ↓
DARE	✓	✓	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
w/o SN	×	✓	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
w/o Cal.	✓	×	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Base	×	×	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]

VI. CONCLUSION

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering. Learned differential rewards remove the component-wise weights, but regularization can still couple the critic’s geometric sensitivity to the score range seen by the policy. DARE treats these as separate design quantities: layer-wise spectral normalization constrains the critic’s sensitivity, and one closed-form calibration sets the score scale without a motion-specific coefficient. The score change relative to the unique zero differential is bounded by the normalized residual, and the critic is then rescaled once, in closed form, to set its operating scale relative to the response width of the reward transform. Across the six skills, DARE attains the lowest macro-average for all four geometric errors, the foot-progress ratio closest to one, and the smallest lateral drift in Table II. The formulation also provides a direct path toward multi-clip training and other morphologies, where additional objectives can be incorporated directly into the differential.

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